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科目：概率論與隨機過程

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1. (a) Denote the number of hours Nat waits for Pat to arrive is W

~~When $0<X<1$, ... f~~

$$F_W(\omega)=P(W\le\omega)=\begin{cases}P(0\le X<1), & \omega=0\\ P(0\le X\le 1+\omega), & 0<\omega\le 1\\ 1 & \omega>1\\ 0, & \omega<0\end{cases}$$

$$=\begin{cases}\frac{1}{2}, & \omega=0\\ \frac{1+\omega}{2}, & \omega\in(0,1]\\ 1, & \omega>1\\ 0, & \omega<0\end{cases}$$

$$f_W(w)=\begin{cases}\frac{1}{2}\delta(w), & w=0\\ \frac{1}{2}, & 0<w\le 1\\ 0, & \text{otherwise.}\end{cases},\quad P(W=0)=\frac{1}{2}$$

$$E[W]=\int_0^1\frac{1}{2}\cdot w\,dw=\frac{1}{4}$$

(b) Denote the ~~expected~~ duration of any particular date ~~of~~ is D.

~~$E[D|W]=$~~

$$E[D|W]=\begin{cases}3, & \cancel{0\le W<1}\ W=0\\ \frac{2-W}{2} & 0<W\le 1\\ \text{undefine}, & W>1.\end{cases}$$

$$E[D]=E_W[E[D|W]]=E[D|W=0]\,P(W=0)+\int_0^1\frac{2-W}{2}\cdot\frac{1}{2}\,dW$$

$$=3\times\frac{1}{2}+\frac{1}{8}\left[2^2-1^2\right]$$

$$=\frac{3}{2}+\frac{3}{8}=\frac{15}{8}$$

(c) Denote the number of dates they will have before breaking up is N

$$P_N(n)=\begin{cases}0, & n\le 0\\ P(0\le W\le\frac{3}{4})^{n-1}\cdot P(W>\frac{3}{4})^2, & n\ge 1\end{cases}=\begin{cases}0, & n\le 0\\ P(0\le X\le 1\frac{3}{4})^{n-1}\cdot P(X>1\frac{3}{4})^2, & n\ge 1\end{cases}$$

$$=\begin{cases}0, & n\le 0\\ \cancel{(\frac{7}{8})^{n-1}\cdot(\frac{1}{8})^2, \quad n\ge 1}\end{cases}$$

$$E[N] = \sum_{n=1}^{\infty} n \cdot \left(\frac{7}{8}\right)^{n-1} \cdot \left(\frac{1}{8}\right)^{2} = \sum_{n=1}^{\infty} (n-1) \cdot \left(\frac{7}{8}\right)^{n-1} \cdot \left(\frac{1}{8}\right)^{2} + \sum_{n=1}^{\infty} \left(\frac{7}{8}\right)^{n-1} \cdot \left(\frac{1}{8}\right)^{2}$$

$$= \sum_{n=1}^{\infty} n \left(\frac{7}{8}\right)^{n} \cdot \left(\frac{1}{8}\right)^{2} + \left(\frac{1}{8}\right)^{2} \cdot \frac{1}{1-\frac{7}{8}}$$

$$= \frac{7}{8} E[N] + \frac{1}{8}$$

$$\Rightarrow E[N] = 1$$

2. Denote the number of different types of pizzas ordered is N when $m \le \min(n,K)-1$ and $K \ge 1$ and $m > 0$.

$$P(N=m \mid K) = P(N \le m \mid K) - P(N \le m-1 \mid K)$$

$$= \frac{\binom{n}{m} \cdot m^K}{n^K} - \frac{\binom{n}{m-1}(m-1)^K}{n^K} = \binom{n}{m}\left(\frac{m}{n}\right)^K - \binom{n}{m-1}\left(\frac{m-1}{n}\right)^k$$

$$P(N=\min(n,K) \mid K) = 1 - P(N \le \min(n,k)-1 \mid K)$$

$$= 1 - \binom{n}{\min(n,k)-1}\left(\frac{\min(n,k)-1}{n}\right)^K$$

$$E[N \mid K] = \sum_{m=1}^{\min(n,k)-1} m P(N=m \mid K) + \min(n,k)\left[1 - \binom{n}{\min(n,K)-1}\left(\frac{\min(n,k)-1}{n}\right)^K\right]$$

$$= \min(n,k) - \sum_{m=1}^{\min(n,k)} \binom{n}{m-1}\left(\frac{m-1}{n}\right)^K$$

$$\cancel{E[N] = E[E[N \mid K]] = \sum_{k=1}^{\infty} E[N \mid K=k]\, p_K(k) = M_K(f(s))}$$

$$\Rightarrow e^{f(s)k} = E[N \mid K=k] = \min(n,k) - \sum_{m=1}^{\min(n,k)} \binom{n}{m-1}\left(\frac{m-1}{n}\right)^k$$

$$f(s) = \frac{\ln\left[\min(n,k) - \sum_{m=1}^{\min(n,k)} \binom{n}{m-1}\left(\frac{m-1}{n}\right)^K\right]}{K}$$

3. $\operatorname{cov}(X,Y) = E[XY] - E[X]E[Y]$

$$E[XY] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xy\, f_{X,Y}(x,y)\,dx\,dy = \frac{\partial^2}{\partial s \partial t} M_{X,Y}(s,t)\Big|_{s=0,\,t=0}$$

$$E[X] = \int_{-\infty}^{+\infty} x f_X(x)\,dx = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} x f_{X,Y}(x,y)\,dx\,dy = \frac{\partial}{\partial s} M_{X,Y}(s,0)\Big|_{s=0}$$

$$E[Y] = \frac{\partial}{\partial t} M_{X,Y}(0,t)\Big|_{t=0}$$

$$\Rightarrow \operatorname{cov}(X,Y) = \frac{\partial^2 M_{X,Y}(s,t)}{\partial s \partial t}\Big|_{(s,t)=(0,0)} - \left(\frac{\partial M_{X,Y}(s,0)}{\partial s}\Big|_{s=0}\right)\left(\frac{\partial M_{X,Y}(0,t)}{\partial t}\Big|_{t=0}\right)$$

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4. (a)

$$M_{n+1} = M_n + \left(1 - \frac{M_n}{a+b}\right) \times 1 = M_n\left(1 - \frac{1}{a+b}\right) + 1$$

(b) $\cancel{M_{n+1} + (a+b) =}$

$$M_{n+1} = M_n\left(1 - \frac{1}{a+b}\right) + 1$$

$$M_{n+1} - (a+b) = \left(M_n - (a+b)\right)\left(1 - \frac{1}{a+b}\right)$$

$$\Rightarrow M_n - (a+b) = \left(M_0 - (a+b)\right)\left(1 - \frac{1}{a+b}\right)^n$$

$$= -b\left(1 - \frac{1}{a+b}\right)^n$$

$$\Rightarrow M_n = a+b - b\left(1 - \frac{1}{a+b}\right)^n$$

(c) Denote the probability that there are k white balls in the urns after ~~the~~ ~~(n+1)th~~ the nth ball is drawn is $P_{n,k}$

$$P(\text{the } (n+1)\text{th ball drawn is white}) = \sum_{k=1}^{\infty} P_{n,k} \cdot \frac{k}{a+b} = \frac{1}{a+b}\sum_{k=1}^{\infty} k P_{n,k} = \frac{M_n}{a+b}$$